Configuring NAT to Conserve Public IP Addresses

Difficulty (out of 3 stars): ★★★ Recommended study time: 1 hour

Contents

1 About This Lab

1.1 Task Description

Since packets with private IP addresses cannot be routed on the Internet, IP packets destined for the Internet cannot reach the egress device of the private network due to lack of routing information.

If a host that uses a private IP address needs to access the Internet, Network Address Translation (NAT) must be configured on the network egress device to translate the private source address of IP data packets into a public source address.

By completing this task, you will gain a clear understanding of local NAT technology and will be able to configure NAT.

1.2 Objectives

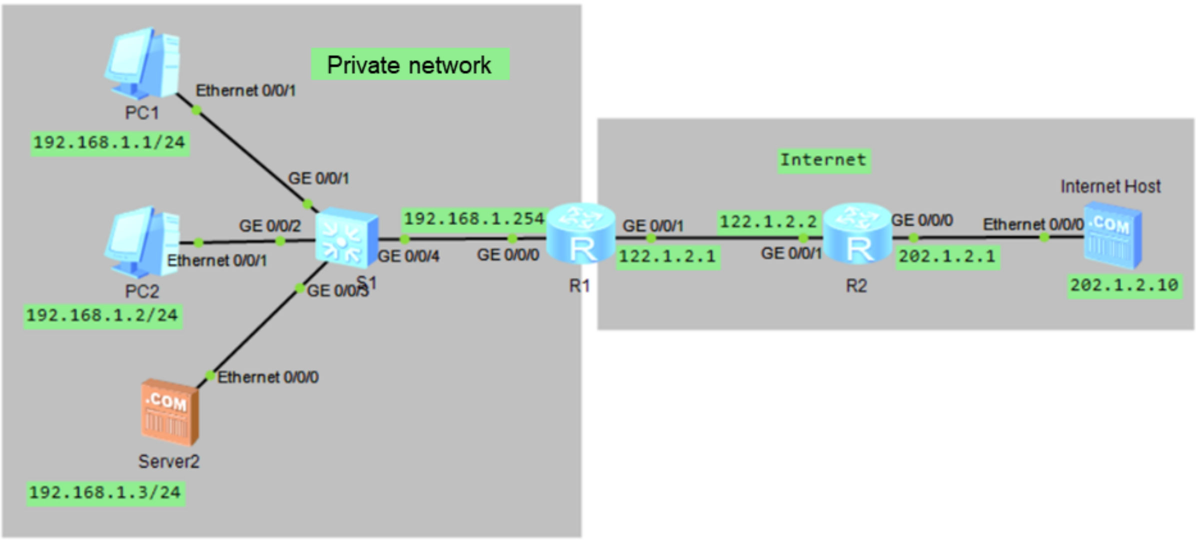
On completing this task, you will be able to:

1. Learn how to configure static NAT.
2. Learn how to configure dynamic NAT.
3. Learn how to configure Network Address and Port Translation (NAPT) and Easy IP.
4. Learn how to configure NAT Server.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Configuring NAT to conserve public IP addresses



2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]ip address 192.168.1.254 24
[R1-GigabitEthernet0/0/0]quit
[R1]
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip address 122.1.2.1 24
[R1-GigabitEthernet0/0/1]quit
[R1]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int g0/0/1
[R2-GigabitEthernet0/0/1]ip address 122.1.2.2 24
[R2-GigabitEthernet0/0/1]quit
[R2]int g0/0/0
[R2-GigabitEthernet0/0/0]ip address 202.1.2.1 24
[R2-GigabitEthernet0/0/0]quit
[R2]
```

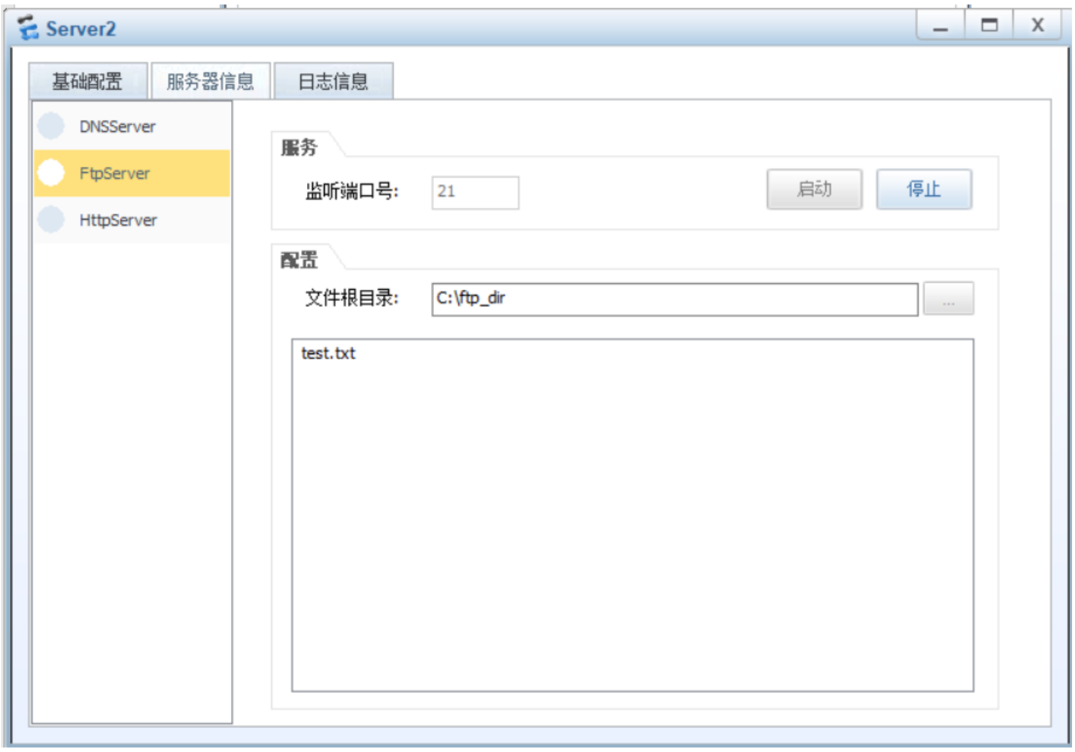
- Configure IP addresses for PC1 and PC2. Figure 2-2 uses PC1 as an example.

Figure 2-2 Configuring an IP address for PC1

- Configure an IP address for the internal server, and enable the FTP service on the server, as shown in Figure 2-3 and Figure 2-4.

Figure 2-3 Configuring an IP address for the internal server

Figure 2-4 Configuring FTP on the internal server



- Configure an IP address for the Internet host, as shown in Figure 2-5.

Figure 2-5 Configuring an IP address for the Internet host

3 Task Implementation

3.1 Checking Basic Configurations

- Run the **ping** command on R1 to check network connectivity.

```
<R1>ping -c 1 192.168.1.1
  PING 192.168.1.1: 56   data bytes, press CTRL_C to break
    Reply from 192.168.1.1: bytes=56 Sequence=1 ttl=128 time=80 ms

<R1>ping -c 1 192.168.1.2
  PING 192.168.1.2: 56   data bytes, press CTRL_C to break
    Reply from 192.168.1.2: bytes=56 Sequence=1 ttl=128 time=100 ms

<R1>ping -c 1 192.168.1.3
  PING 192.168.1.3: 56   data bytes, press CTRL_C to break
    Reply from 192.168.1.3: bytes=56 Sequence=1 ttl=128 time=80 ms

<R1>ping -c 1 122.1.2.2
  PING 122.1.2.2: 56   data bytes, press CTRL_C to break
    Reply from 122.1.2.2: bytes=56 Sequence=1 ttl=255 time=60 ms
<R1>
```

- Run the **ping** command on R2 to check network connectivity.

```
<R2>ping -c 2 202.1.2.10
```

```
  PING 202.1.2.10: 56   data bytes, press CTRL_C to break
    Reply from 202.1.2.10: bytes=56 Sequence=1 ttl=255 time=40 ms
    Reply from 202.1.2.10: bytes=56 Sequence=2 ttl=255 time=10 ms

<R2>ping -c 2 122.1.2.1
  PING 122.1.2.1: 56   data bytes, press CTRL_C to break
    Reply from 122.1.2.1: bytes=56 Sequence=1 ttl=255 time=20 ms
    Reply from 122.1.2.1: bytes=56 Sequence=2 ttl=255 time=30 ms
<R2>
```

3.2 Configuring R1 to Access the Internet Host

Configure a static route on R1 so that it can access the Internet host.

```
[R1]ip route-static 0.0.0.0 0.0.0.0 122.1.2.2
[R1]
[R1]ping 202.1.2.10
  PING 202.1.2.10: 56   data bytes, press CTRL_C to break
    Reply from 202.1.2.10: bytes=56 Sequence=1 ttl=254 time=50 ms
    Reply from 202.1.2.10: bytes=56 Sequence=2 ttl=254 time=20 ms
    Reply from 202.1.2.10: bytes=56 Sequence=3 ttl=254 time=20 ms
    Reply from 202.1.2.10: bytes=56 Sequence=4 ttl=254 time=30 ms
    Reply from 202.1.2.10: bytes=56 Sequence=5 ttl=254 time=20 ms

  --- 202.1.2.10 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/28/50 ms

[R1]
```

3.3 Configuring Static NAT

Static NAT: A private IP address is mapped to a fixed public IP address. That is, there is a one-to-one mapping between private and public IP addresses.

1. Configure static NAT on R1 to map private addresses of internal hosts to public addresses in one-to-one mode.

   Run the following commands on G0/0/1 of R1:

   ```
   [R1]int g0/0/1
   [R1-GigabitEthernet0/0/1]nat static global 122.1.2.3 inside 192.168.1.1
   [R1-GigabitEthernet0/0/1]nat static global 122.1.2.4 inside 192.168.1.2
   [R1-GigabitEthernet0/0/1]nat static global 122.1.2.5 inside 192.168.1.3
   [R1-GigabitEthernet0/0/1]quit
   [R1]
   ```

2. On PC1, ping the Internet host address 202.1.2.10.

   ```
   PC>ping 202.1.2.10

   Ping 202.1.2.10: 32 data bytes, Press Ctrl_C to break
   ```

```
From 202.1.2.10: bytes=32 seq=1 ttl=255 time=32 ms
From 202.1.2.10: bytes=32 seq=2 ttl=255 time=47 ms
From 202.1.2.10: bytes=32 seq=3 ttl=255 time=31 ms
From 202.1.2.10: bytes=32 seq=4 ttl=255 time=31 ms
From 202.1.2.10: bytes=32 seq=5 ttl=255 time=31 ms

--- 202.1.2.10 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 31/34/47 ms

PC>ping 202.1.2.10
<R2>
```

3.4 Configuring Dynamic NAT

Static NAT enforces a strict one-to-one mapping between private and public IP addresses. Even when an internal host is offline or inactive for a long time, its corresponding public IP remains occupied. To avoid wasting addresses, dynamic NAT introduces the concept of an address pool. All available public IP addresses form this pool.

1. Delete the static NAT configuration on G0/0/1 of R1.

```
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]undo nat static global 122.1.2.3 inside 192.168.1.1
[R1-GigabitEthernet0/0/1]undo nat static global 122.1.2.4 inside 192.168.1.2
[R1-GigabitEthernet0/0/1]undo nat static global 122.1.2.5 inside 192.168.1.3
[R1-GigabitEthernet0/0/1]quit
[R1]
```

2. Configure dynamic NAT on R1 to dynamically map private addresses of internal hosts to public addresses.

   - Create an address pool.

```
[R1]nat address-group 1 122.1.2.3 122.1.2.4
[R1]
```

   - Configure an ACL rule for NAT.

     Configure a basic ACL to match the source address range that requires dynamic NAT.

```
[R1]acl 2000
[R1-acl-basic-2000]rule 5 permit source 192.168.1.0 0.0.0.255
[R1-acl-basic-2000]quit
[R1]
```

   - On R1, configure dynamic NAT on the interface connected to the Internet to dynamically map private addresses of internal hosts to public addresses. The **no-pat** parameter indicates that port translation is not performed.

```
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]nat outbound 2000 address-group 1 no-pat
[R1-GigabitEthernet0/0/1]quit
[R1]
```

   - On PC1, ping the Internet host address 202.1.2.10.

```
PC>ping 202.1.2.10
```

```
Ping 202.1.2.10: 32 data bytes, Press Ctrl_C to break
From 202.1.2.10: bytes=32 seq=1 ttl=255 time=31 ms
From 202.1.2.10: bytes=32 seq=2 ttl=255 time=31 ms
From 202.1.2.10: bytes=32 seq=3 ttl=255 time=16 ms
From 202.1.2.10: bytes=32 seq=4 ttl=255 time=62 ms
From 202.1.2.10: bytes=32 seq=5 ttl=255 time=32 ms

--- 202.1.2.10 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 16/34/62 ms

PC>
```

3.5 Configuring Easy IP

When dynamic NAT selects an address from the address pool for translation, it does not modify the port number. This is known as No-Port Address Translation (No-PAT). In this case, the public and private addresses still follow a 1:1 mapping, and the utilization of public IP addresses is not significantly improved.

When NAPT selects an address from the address pool for address translation, it not only translates the IP address but also translates the port number, enabling a 1:N mapping between public and private addresses. This greatly improves the utilization of public IP addresses.

Like NAPT, Easy IP translates both IP addresses and transport-layer port numbers. The difference is that Easy IP does not use an address pool; instead, it uses an interface address as the post-NAT public address.

1. Delete the dynamic NAT configuration.
   - Before deleting the address pool created for dynamic NAT, delete the existing NAT configuration from the interface.

```
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]undo nat outbound 2000 address-group 1 no-pat
[R1-GigabitEthernet0/0/1]quit
[R1]undo nat address-group 1
[R1]
```

2. Configure Easy IP.

```
[R1]acl 2000
[R1-acl-basic-2000]rule 5 permit source 192.168.1.0 0.0.0.255
[R1-acl-basic-2000]quit
[R1]
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]nat outbound 2000
[R1-GigabitEthernet0/0/1]quit
[R1]
```

3. On PC1, ping the Internet host address 202.1.2.10.

```
PC>ping 202.1.2.10 -t

Ping 202.1.2.10: 32 data bytes, Press Ctrl_C to break
```

```
From 202.1.2.10: bytes=32 seq=1 ttl=253 time=31 ms
From 202.1.2.10: bytes=32 seq=2 ttl=253 time=47 ms
From 202.1.2.10: bytes=32 seq=3 ttl=253 time=15 ms
From 202.1.2.10: bytes=32 seq=4 ttl=253 time=31 ms
From 202.1.2.10: bytes=32 seq=5 ttl=253 time=47 ms
From 202.1.2.10: bytes=32 seq=6 ttl=253 time=31 ms
From 202.1.2.10: bytes=32 seq=7 ttl=253 time=47 ms
From 202.1.2.10: bytes=32 seq=8 ttl=253 time=63 ms
From 202.1.2.10: bytes=32 seq=9 ttl=253 time=47 ms
From 202.1.2.10: bytes=32 seq=10 ttl=253 time=31 ms
From 202.1.2.10: bytes=32 seq=11 ttl=253 time=31 ms
From 202.1.2.10: bytes=32 seq=12 ttl=253 time=31 ms
```

4. Run the **display nat session all** command on R1 to view the NAT translation table.

```
[R1]display nat session all
  NAT Session Table Information:

     Protocol              : ICMP(1)
     SrcAddr    Vpn        : 192.168.1.1
     DestAddr   Vpn        : 202.1.2.10
     Type Code IcmpId      : 0    8    13550
     NAT-Info
       New SrcAddr         : 122.1.2.1
       New DestAddr        : ----
       New IcmpId          : 10246

     Protocol              : ICMP(1)
     SrcAddr    Vpn        : 192.168.1.1
     DestAddr   Vpn        : 202.1.2.10
     Type Code IcmpId      : 0    8    13549
     NAT-Info
       New SrcAddr         : 122.1.2.1
       New DestAddr        : ----
       New IcmpId          : 10245

  Total : 2
[R1]
```

3.6 Configuring NAT Server

NAT Server defines a one-to-one mapping between a [public IP:port] and a [private IP: port], allowing internal servers to be exposed to the public network. It is used when a server on the private network needs to provide services to external users.

```
[R1]interface GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1]nat server protocol tcp global 122.1.2.100 ftp inside
192.168.1.3 ftp
[R1-GigabitEthernet0/0/1]quit
[R1]quit
```

Run the **ftp** command on R2 to download the **test.txt** file from the FTP server.

```
<R2>
<R2>ftp 122.1.2.100
Trying 122.1.2.100 ...

Press CTRL+K to abort
Connected to 122.1.2.100.
220 FtpServerTry FtpD for free
User(122.1.2.100:(none)):
331 Password required for   .
Enter password:
230 User   logged in , proceed

[R2-ftp]dir
200 Port command okay.
150 Opening ASCII NO-PRINT mode data connection for ls -l.
-rwxrwxrwx   1              nogroup              24 Apr 10   2025 test.txt
226 Transfer finished successfully. Data connection closed.
FTP: 68 byte(s) received in 0.050 second(s) 1.36Kbyte(s)/sec.

[R2-ftp]get test.txt
200 Port command okay.
150 Sending test.txt (24 bytes). Mode STREAM Type BINARY
226 Transfer finished successfully. Data connection closed.
FTP: 24 byte(s) received in 0.250 second(s) 96.00byte(s)/sec.

[R2-ftp]
[R2-ftp]bye
221 Goodbye.

<R2>
```

4 Verification

Omitted

5 Quiz

1. Which types of NAT translation allow external network devices to proactively access internal servers?

   Both static NAT and NAT Server can achieve this. Static NAT enables bidirectional communication, naturally allowing external network devices to access internal servers. NAT Server is specifically designed to allow external network devices to initiate connections to internal servers.

2. What are the advantages of NAPT compared to No-PAT?

   NAPT can translate multiple private IP addresses into one public IP address, improving public IP address utilization.

3. What are the differences between NAPT and Easy IP?